Test of the overarrows package compatibility with LaTeX engines

Julien Labbé

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Predefined commands

```
\begin{align*}
  \vv{AB}                    &  & \vec{v}                    \\
  \overrightarrow{AB}        &  & \underrightarrow{AB}       \\
  \overleftarrow{AB}         &  & \underleftarrow{AB}        \\
  \overleftrightarrow{AB}    &  & \underleftrightarrow{AB}   \\
  \overrightharpoonup{AB}    &  & \underrightharpoonup{AB}   \\
  \overrightharpoondown{AB}  &  & \underrightharpoondown{AB} \\
  \overleftharpoonup{AB}     &  & \underleftharpoonup{AB}    \\
  \overleftharpoondown{AB}   &  & \underleftharpoondown{AB}  \\
  \overbar{AB}               &  & \underbar{AB}
\end{align*}
```

$$\begin{aligned}
\overrightarrow{\overrightarrow{AB}} & & \vec{v}   \\
\overrightarrow{AB}   & & \underrightarrow{AB}   \\
\overleftarrow{AB}   & & \underleftarrow{AB}   \\
\overleftrightarrow{AB}   & & \underleftrightarrow{AB}   \\
\overrightharpoonup{AB}   & & \underrightharpoonup{AB}   \\
\overrightharpoondown{AB}   & & \underrightharpoondown{AB} \\
\overleftharpoonup{AB}   & & \underleftharpoonup{AB}   \\
\overleftharpoondown{AB}   & & \underleftharpoondown{AB} \\
\overline{AB}   & & \underline{AB}
\end{aligned}$$

harrowextender

```
\begin{align*}
  \mathtt{-} ~=~                                        & -                                  \\
  \mathtt{\backslash relbar} ~=~                        & \relbar                            \\
  \mathtt{\backslash relbareda} ~=~                     & \relbareda                         \\
  \mathtt{\backslash mathrel \backslash harrowextender} ~=~ & \mathrel{\harrowextender} \\
  \mathtt{\backslash ovar@auto@middle} ~=~              & \UseName{ovar@auto@middle}
\end{align*}
```

$$
\begin{aligned}
- &= - \\
\mathtt{\backslash relbar} &= - \\
\mathtt{\backslash relbareda} &= \text{-} \\
\mathtt{\backslash mathrel\backslash harrowextender} &= \\
\mathtt{\backslash ovar@auto@middle} &= \text{-}
\end{aligned}
$$

```
\NewOverArrowCommand{\testharrowextender}{%
  middle config=auto
}
\TestOverArrow{\testharrowextender}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

```
\NewOverArrowCommand{\useharrowextender}{%
  middle config=harrowextender
}
\TestOverArrow{\useharrowextender}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

TikZ arrows

```
\NewOverArrowCommand[tikz]{\tikzstandardthickness}{%
  path options={
    x=\overarrowlength, line width=\overarrowthickness
  },
  path={(0,0) -- (1,0)}, arrows={->}
}
\NewOverArrowCommand[tikz]{\tikzsmallerthickness}{%
  path options={
    x=\overarrowlength, line width=\overarrowsmallerthickness,
  },
  path={(0,0) -- (1,0)}, arrows={->}
}
\begin{align*}
  \tikzstandardthickness{ABCD} & &
  \scriptscriptstyle \tikzstandardthickness{ABCD} \\
  \tikzsmallerthickness{ABCD} & &
  \scriptscriptstyle \tikzsmallerthickness{ABCD}
\end{align*}
```

$$\overrightarrow{ABCD} \qquad \overrightarrow{_{ABCD}}$$

$$\overrightarrow{ABCD} \qquad \overrightarrow{_{ABCD}}$$